

Assignment - 04.

Q1.

→

LIST	TUPLE
Mutable	Immutable.
Syntax - []	Syntax - ()
Uses more Memory	Uses less memory
Slow	Faster perf..

Q2.

→

Tuples are faster because they are immutable, so PVM do not need to allocate extra memory for modification or maintain methods that change the object.

Advantages of tuple.

- i] Faster access.
- ii] Less memory usage.
- iii] Safer for fixed data

Q3.

→

lst[0] = 100 → works.

tpl[0] = 100 → Wrong syntax. [tuples are not mutable]

Output: Error

Q4.

→

Strings are immutable to improve:

- Memory efficiency.
- Security
- Performance

So, `s1 = "Python"`
`s1 = s1 + "2026"`

→ here A new string `s1`, is created & now variable `s1` refers to the new object.

Q5.

→ Output - Different memory address.

Q6.

→ A dictionary is a collection of key-value pairs

Syntax:

```
student = {  
    "Name" : "Athu",  
    "Age" : 20,  
}
```

here Name, Age are keys.

Ex `student["Name"]` → Athu

Q7.

→ `{1: "ONE", 2: "TWO"}`

Q8.

→ `range()` generates a sequence of numbers, It is commonly used in loops.

eg. `r = range(5)` → Stops one step before.
→ 0, 1, 2, 3, 4, ~~5~~

Q 9.

→ Output - $\text{range}(0, 5)$ start. → end. →
 $[0, 1, 2, 3, 4]$

Q 10.

→ $\text{range}(1, 10)$ vs $\text{range}(1, 10, 2)$

here, the only difference is of increment value,
in first range the value increases by 1 all upto
9.

while in 2nd range, the value increases by 2,
like - 1, 3, 5, 7, 9